Hengfa Lu, Ph.D.

Postdoctoral Fellow, Department of Biomedical Engineering
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Research Interests

I develop computational biomedical imaging systems through the joint design of measurement operators, optical and electronic instrumentation, and physics constrained inference. My research integrates coherent optical sensing, signal processing, inverse problems, and machine learning, building on earlier work in MRI and NMR reconstruction. My independent program aims to advance adaptive acquisition and fast, affordable wide-field quantitative imaging for investigating tissue function and recovery.

Academic and Research Appointments

Postdoctoral Fellow Feb. 2026 – Present
Department of Biomedical Engineering, The University of Texas at Austin *Austin, Texas*
Mentor: Andrew K. Dunn.

Research Assistant Jun. 2018 – Feb. 2020
Department of Electronic Science, Xiamen University *Xiamen, China*
Mentor: Xiaobo Qu.

Education

Ph.D., Biomedical Engineering Jan. 2021 – Dec. 2025
The University of Texas at Austin *Austin, Texas*
Advisor: Andrew K. Dunn.
Dissertation: Modulated Illumination Measurement and Self Supervised Physics Informed Inference

M.E., Electronics and Communication Engineering Sept. 2015 – Jun. 2018
Xiamen University *Xiamen, China*

B.E., Electronic Engineering Sept. 2011 – Jun. 2015
Shantou University *Shantou, China*

Peer-Reviewed Journal Articles

J1. **H. Lu**, Q. Fang, J. A. Ashbrook, V. Nemchek, M. Fracassi, T. A. Jones, and A. K. Dunn, Mapping fast tissue dynamics with long camera exposures via intensity modulation, *Proceedings of the National Academy of Sciences*, vol. 123, no. 17, e2524940123, 2026.

J2. X. Zhang, **H. Lu**, Y. Huang, T. Tan, X. Zhang, and T. Tong, Accelerating Hankel low-rank MR reconstruction with a patch-based Hankel scheme, *Biomedical Signal Processing and Control*, vol. 115, 109426, 2026.

J3. X. Zhang, **H. Lu**, D. Guo, Z. Lai, H. Ye, X. Peng, B. Zhao, and X. Qu, Accelerated MRI reconstruction with separable and enhanced low-rank Hankel regularization, *IEEE Transactions on Medical Imaging*, vol. 41, no. 9, pp. 2486–2498, 2022.

J4. E. Scope Crafts, **H. Lu**, H. Ye, L. L. Wald, and B. Zhao, An efficient approach to optimal experimental design for magnetic resonance fingerprinting with B-splines, *Magnetic Resonance in Medicine*, vol. 88, no. 1, pp. 239–253, 2022.

J5. X. Zhang, **H. Lu**, D. Guo, L. Bao, F. Huang, Q. Xu, and X. Qu, A guaranteed convergence analysis for the projected fast iterative soft-thresholding algorithm in parallel MRI, *Medical Image Analysis*, vol. 69, 101987, 2021.

J6. Y. Pustovalova, F. Delaglio, D. L. Craft, H. Arthanari, A. Bax, M. Billeter, M. J. Bostock, H. Dashti, D. F. Hansen, S. G. Hyberts, B. A. Johnson, K. Kazimierczuk, **H. Lu**, M. Maciejewski, T. M. Miljenović, M. Mobli, D. Nietlispach, V. Orekhov, R. Powers, X. Qu, S. A. Robson, D. Rovnyak, G. Wagner, J. Ying, M. Zambrello, J. C. Hoch, D. L. Donoho, and A. D. Schuyler, NUScon: A community-driven platform for quantitative evaluation of nonuniform sampling in NMR, *Magnetic Resonance*, vol. 2, no. 2, pp. 843–861, 2021.

Peer-Reviewed Journal Articles (continued)

J7. X. Qu, Y. Huang, **H. Lu**, T. Qiu, D. Guo, T. Agback, V. Orekhov, and Z. Chen, Accelerated nuclear magnetic resonance spectroscopy with deep learning, *Angewandte Chemie International Edition*, vol. 59, no. 26, pp. 10297–10300, 2020.

J8. **H. Lu**, X. Zhang, T. Qiu, J. Yang, D. Guo, Z. Chen, and X. Qu, Low rank enhanced matrix recovery of hybrid time and frequency data in fast magnetic resonance spectroscopy, *IEEE Transactions on Biomedical Engineering*, vol. 65, no. 4, pp. 809–820, 2018.

J9. X. Qu, T. Qiu, D. Guo, **H. Lu**, J. Ying, M. Shen, B. Hu, V. Orekhov, and Z. Chen, High-fidelity spectroscopy reconstruction in accelerated NMR, *Chemical Communications*, vol. 54, no. 78, pp. 10958–10961, 2018.

J10. D. Guo, **H. Lu**, and X. Qu, A fast low rank Hankel matrix factorization reconstruction method for non-uniformly sampled magnetic resonance spectroscopy, *IEEE Access*, vol. 5, pp. 16033–16039, 2017.

J11. J. Ying, **H. Lu**, Q. Wei, J.-F. Cai, D. Guo, J. Wu, Z. Chen, and X. Qu, Hankel matrix nuclear norm regularized tensor completion for *N*-dimensional exponential signals, *IEEE Transactions on Signal Processing*, vol. 65, no. 14, pp. 3702–3717, 2017.

J12. Z. Lai, X. Qu, **H. Lu**, X. Peng, D. Guo, Y. Yang, G. Guo, and Z. Chen, Sparse MRI reconstruction using multi-contrast image guided graph representation, *Magnetic Resonance Imaging*, vol. 43, pp. 95–104, 2017.

Preprints

P1. **H. Lu**, J. A. Ashbrook, and A. K. Dunn, Label-free, physics-constrained learning for multi-exposure speckle imaging parameter estimation, *bioRxiv*, 2026. Manuscript under revision for *IEEE Transactions on Medical Imaging*.

P2. **H. Lu**, L. Parker, J. W. Tunnell, and A. K. Dunn, Sinusoidal intensity modulation speckle imaging for diffuse flow measurements, *Biomedical Optics Express*, 2026. (Plan to submit October 2026).

P3. **H. Lu**, H. Ye, L. L. Wald, and B. Zhao, Accelerated MR fingerprinting with low-rank and generative subspace modeling, *arXiv:2305.10651 [eess.IV]*, 2023. (Preliminary version published in *IEEE ICASSP Workshops*, 2023.)

Peer-Reviewed Full Conference Papers

C1. Q. Fang, **H. Lu**, A. Tomar, and A. K. Dunn, Intensity-modulated wide-field imaging for characterizing intensity dynamics, in *Proceedings of SPIE 13320, Biomedical Light Scattering XV*, 133200B, 2025.

C2. **H. Lu** and X. Zhang, A fast patch-based Hankel low-rank method for magnetic resonance spectroscopy reconstruction, in *Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1–4, 2024.

C3. **H. Lu**, A. Sun, J. Wang, and B. Zhao, Accelerated phase-contrast flow MRI with low-rank and generative subspace modeling, in *Proceedings of the 57th Asilomar Conference on Signals, Systems, and Computers*, pp. 27–31, 2023.

C4. **H. Lu** and B. Zhao, Accelerated magnetic resonance parameter mapping with low-rank modeling and deep generative priors, in *Proceedings of the IEEE Statistical Signal Processing Workshop*, pp. 403–407, 2023.

C5. **H. Lu** and B. Zhao, Accelerated magnetic resonance fingerprinting with low-rank and generative subspace modeling, in *Proceedings of IEEE ICASSP Workshops*, pp. 1–5, 2023.

C6. A. Sun, **H. Lu**, P. Wu, and B. Zhao, Accelerated black-blood cine MR imaging with low-rank and sparsity constraints, in *Proceedings of IEEE EMBC*, pp. 1–4, 2023.

C7. **H. Lu**, H. Ye, and B. Zhao, Improved balanced steady-state free precession based MR fingerprinting with deep autoencoders, in *Proceedings of IEEE EMBC*, pp. 3029–3034, 2022.

C8. **H. Lu**, X. Zhang, T. Qiu, J. Yang, D. Guo, Z. Chen, and X. Qu, A low rank Hankel matrix reconstruction method for ultrafast magnetic resonance spectroscopy, in *Proceedings of IEEE EMBC*, pp. 3269–3272, 2017.

Selected Honors and Awards

IEEE EMBC NextGen Scholar Award, IEEE Engineering in Medicine and Biology Society 2024
Summa Cum Laude Merit Award (top 5%), ISMRM & ISMRT Annual Meeting 2023
Cockrell School of Engineering Fellowship, The University of Texas at Austin 2021
Outstanding Master's Thesis Award, Xiamen University 2018
National Scholarship for Graduate Students 2017

Research Mentoring

Graduate Research Mentoring, The University of Texas at Austin

Provided research and technical guidance to graduate students in computational biomedical imaging, including inverse problem formulation, signal processing, algorithm development, and experimental validation.

Undergraduate Research Mentoring, The University of Texas at Austin

Rishik Boddeti, Undergraduate Researcher, Electrical and Computer Engineering May 2025 – Present
Project: Development of compact acquisition and control hardware for speckle imaging, including circuit and PCB design, hardware software integration, and experimental validation.

Luke Underhill, Undergraduate Researcher, Physics Jan. 2026 – Present
Project: Deep learning methods for image denoising in laser speckle contrast imaging.

Teaching Experience and Interests

Teaching Assistant, *Biomedical Instrumentation* (undergraduate) Spring 2024
The University of Texas at Austin *Austin, Texas*
Held office hours, supported laboratory exercises and course projects, and graded assignments. Mean student ratings ranged from 4.50 to 4.68 out of 5 across questions assessing availability, approachability, clarity, and engagement.

Teaching Assistant, *Circuits Analysis* (undergraduate) Spring 2016
Xiamen University *Xiamen, China*
Held office hours and graded coursework.

Teaching interests: Signals and Systems; Digital and Statistical Signal Processing; Image Processing; Computational Imaging; Numerical Methods; Biomedical Instrumentation; Biomedical Optics.

Professional Service

Session Chair, Image Reconstruction and Enhancement, IEEE EMBC 2022

Journal Reviewer: *IEEE Transactions on Medical Imaging*; *IEEE Transactions on Computational Imaging*; *IEEE Transactions on Instrumentation and Measurement*; *Pattern Recognition*; *Neurocomputing*; *Scientific Reports*; *Magnetic Resonance Imaging*; *Computerized Medical Imaging and Graphics*; *Biomedical Signal Processing and Control*; *Optics & Laser Technology*.

Conference Reviewer: International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026; IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2026; IEEE International Symposium on Biomedical Imaging (ISBI), 2024 – 2026; IEEE International Conference on Image Processing (ICIP), 2022 – 2026; Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2022.